Olympiad Test Paper — Science (Class 7) — Paper 3

Chapter 9: Soil

Paper 3 — (progressively harder)

Subject	Science (NCERT Class 7)
Chapter	9 — Soil
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: **+4** for each correct answer · **−1** for each wrong answer · **0** if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Watch for traps that swap percolation rate with water-holding capacity, confuse which horizon gains vs loses washed-down minerals, and assume 'more water always helps the crop' regardless of aeration. Do not write on this paper — mark answers on your answer sheet.

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1. A field study finds, top to bottom: a thin spongy layer of half-rotted leaves, then a dark crumbly band, then a paler band where the dark band's washed-down clay and iron have piled up, then loose rock fragments, then solid rock. Reading top to bottom, which sequence of horizon names is correct?

(A) O, B, A, C, bedrock (B) A, O, C, B, bedrock
(C) O, A, B, C, bedrock (D) bedrock, C, B, A, O

2. In a soil profile the B horizon is often called the 'zone of accumulation'. The fine clay and dissolved minerals that collect there mostly arrive by:

(A) rising up from the bedrock by capillary action
(B) wind blowing them in sideways from neighbouring fields
(C) rainwater carrying them downward from the A horizon above
(D) earthworms carrying them up from the C horizon

3. Two adjacent plots have the same climate. Plot 1 has been forest for centuries; plot 2 was cleared and farmed bare for decades. The forest plot's A horizon is much thicker and darker. The best explanation is that the forest plot:

(A) received a steady supply of leaf litter that decayed into humus and was not stripped by erosion

(B) had faster weathering of bedrock because trees crack rock

(C) had a thicker B horizon that pushed the A horizon upward

(D) drained water faster, so more minerals stayed in the topsoil

4. A student claims, 'The C horizon is just smaller, looser pieces of the bedrock right below it.' Judging this claim against how soil forms, it is:

(A) wrong, because the C horizon is the humus-rich layer

(B) broadly correct, because C is partly weathered parent rock and little affected by humus

(C) wrong, because the C horizon lies above the topsoil

(D) wrong, because the C horizon is made only of living organisms

5. On a freshly exposed lava flow, scientists return after 50, 500 and 5000 years and measure topsoil depth: almost none, a thin layer, then a deeper layer. The pattern best illustrates that:

(A) lava turns into deep fertile soil within the first few rainstorms

(B) soil depth depends only on how much it rained that year

(C) the deepest soil forms first and then gets thinner with time

(D) soil builds up extremely slowly as rock weathers and organisms add humus over long times

6. Which combination of agents working over long time periods best accounts for the breakdown of solid rock into soil parent material?

(A) freezing and thawing water, wind abrasion, and growing plant roots prying rock apart

(B) a single earthquake instantly grinding rock to soil

(C) the weight of the bedrock alone crushing itself to powder

(D) humus chemically rebuilding rock from decayed leaves

7. A hot, very wet tropical hillside often has deep, heavily weathered soil, while a cold dry mountaintop has almost bare rock. The main reason the tropical site weathers faster is that:

(A) cold dry air dissolves rock faster than warm wet air

(B) tropical rock is always softer than mountain rock

(C) high altitude automatically produces deeper soil

(D) warmth and abundant water speed up the chemical and physical breakdown of rock

8. Soil is sometimes described as having four main parts. If a sample were truly missing the air component, the most direct consequence for plant roots would be that they:

(A) would lose all access to mineral nutrients

(B) could not get enough oxygen and would struggle to breathe

(C) would be unable to take up any water at all

(D) would immediately convert the soil to humus

9. Three samples have these particle descriptions: P = mostly tiny tightly-packed particles; Q = mostly large loose grains; R = a balanced mix of large, medium and tiny particles with humus. Matching to soil types, P, Q and R are respectively:

(A) sandy, clayey, loamy (B) loamy, clayey, sandy

(C) clayey, sandy, loamy (D) clayey, loamy, sandy

10. A student writes: 'Clayey soil percolates slowly because its particles are large, leaving big gaps that trap water.' The reasoning is faulty because in clayey soil the particles are actually:

(A) very large with huge gaps, which is why water drains fast

(B) very small and tightly packed, leaving tiny pores that water moves through only slowly

(C) all exactly the same size as sand grains

(D) so loose that the soil cannot hold any water

11. Two pots hold equal volumes of soil. Pot A is sandy; pot B is clayey. The same amount of water is poured on each. Compared with pot B, pot A will end up with:

(A) more water draining out the bottom and less retained in the soil

(B) more water retained in the soil and less draining out

(C) exactly the same drainage and retention as pot B

(D) no drainage at all because sand blocks water

12. A gardener says, 'I want the soil with the highest water-holding capacity, so I'll also get the fastest percolation.' Why is this expectation self-contradictory?

(A) high water-holding capacity always comes with the fastest percolation, so the wish is fine

(B) high water-holding capacity goes with small tight pores, which give a low (slow) percolation rate

(C) percolation and water-holding capacity are the same property measured twice

(D) only loamy soil can have both the highest retention and the fastest drainage at once

13. Aeration (the amount of air available to roots) is generally poorest in:

(A) loose sandy soil with large air-filled gaps(B) well-drained loamy soil after rain

(C) water-logged clayey soil, whose tiny pores stay filled with water

(D) dry sandy soil on a breezy day

14. A sample of loamy soil is described as 'the all-rounder' for most crops. The single best reason is that it:

(A) drains faster than any other soil, so roots never sit in water

(B) holds more water than any other soil, so crops never need watering

(C) contains only large particles, making it impossible to compact

(D) holds moisture for roots yet still lets surplus water drain and stays well aerated

15. After a storm, three identical garden beds — sandy, loamy and clayey — are checked. Which observation is most consistent with their known properties?

(A) the clayey bed dried first because clay drains fastest

(B) the sandy bed stayed soggy longest because sand holds the most water

(C) all three beds dried at exactly the same time

(D) the sandy bed dried first, the clayey bed stayed soggy longest, the loamy bed was in between

16. Mixing a generous amount of well-rotted humus (compost) into a pure sandy soil will most usefully:

(A) turn it instantly into solid clay with no air

(B) stop it from ever percolating water again

(C) raise its water-holding capacity and add nutrients while keeping reasonable drainage

(D) remove all the soil's organisms so it stays clean

17. Why does a freshly ploughed, crumbly field usually let water soak in better than the same field after it has been packed hard by foot traffic?

(A) ploughing changes the soil's particle sizes to all-large grains

(B) loosening opens up pore spaces between particles, raising percolation

(C) packing the soil adds humus that speeds drainage

(D) compacted soil has bigger pores than loose soil

18. A salt-affected sandy patch and a humus-rich loamy patch are each sampled for living organisms. The loamy patch teems with far more life chiefly because it:

(A) is colder, and cold soil always holds more organisms

(B) offers moisture, humus food and air pockets that organisms need

(C) has the largest particles, which organisms prefer

(D) drains so fast that organisms are never disturbed

19. Paddy (rice) is grown in flooded fields, while groundnut needs loose, well-drained ground. The best soil pairing is therefore:

(A) sandy soil for paddy and clayey soil for groundnut

(B) clayey soil for both paddy and groundnut

(C) clayey soil for paddy and sandy or sandy-loam soil for groundnut

(D) gravel for paddy and water-logged clay for groundnut

20. A farmer on heavy clay wants to grow a crop well suited to soil that stays moist and holds water. The best choice from the list is:

(A) cactus

(B) groundnut

(C) paddy (rice)

(D) watermelon on a sandy ridge

21. Wheat and gram do well on clayey-to-loamy soils, whereas cotton and lentils favour lighter sandy-loam. A farmer with deep, water-retentive clayey-loam should sensibly plant:

(A) cotton, because cotton needs water-logged clay

(B) only crops that need pure sand

(C) wheat or gram, which suit moisture-holding clayey-loam

(D) no crop, because clayey-loam supports nothing

22. A new field has been left as bare, fast-draining sandy-loam. Which crop choice is best matched to it, and why?

(A) lentils, because they tolerate lighter, well-drained sandy-loam

(B) paddy, because rice grows best where water drains away fast

(C) any crop, since soil type makes no difference to crops

(D) only crops that need permanently flooded fields

23. A gardener insists, 'More water is always better for any crop, so I'll keep my clay bed flooded.' For a crop like gram, this is wrong mainly because:

(A) flooding makes the clay drain too fast and dries the roots out

(B) extra water turns the clay into sandy soil

(C) flooding fills the clay's pores with water and starves the roots of air

(D) gram is a water plant that needs permanent flooding

24. Two soils are being chosen for a vegetable garden that needs steady moisture but must not stay water-logged. Soil M holds water moderately and drains the surplus; soil N holds water strongly and drains very slowly. The better choice and reason is:

(A) soil M, because moderate retention with drainage avoids both drying out and water-logging

(B) soil N, because the more water a soil holds the better for every crop

(C) soil N, because slow drainage means roots always get air

(D) soil M, because it holds no water at all and never needs draining

25. Percolation rate is calculated as the volume of water that drains through divided by the time taken. In an experiment, 250 mL of water drains through a sandy column in 5 minutes. The percolation rate is:

(A) 1250 mL/min

(B) 50 mL/min

(C) 0.02 mL/min

(D) 245 mL/min

26. A clayey sample lets 90 mL of water through in 30 minutes. Its percolation rate is:

(A) 2700 mL/min

(B) 0.33 mL/min

(C) 60 mL/min

(D) 3 mL/min

27. Soil X drains 200 mL in 4 minutes; soil Y drains 200 mL in 25 minutes. The ratio of their percolation rates (X : Y) is:

(A) 4 : 25
(B) 1 : 1
(C) 200 : 200
(D) 25 : 4

28. Soil P percolates 300 mL in 6 minutes; soil Q percolates 480 mL in 12 minutes. Which soil has the higher percolation rate?

(A) soil Q, because it let a larger total volume through
(B) soil P, at 50 mL/min versus soil Q's 40 mL/min
(C) they are equal because both drain hundreds of mL
(D) soil Q, because it took the longer time

29. A soil has a percolation rate of 15 mL/min. How long will it take 180 mL of water to percolate through it at this rate?

(A) 2700 minutes
(B) 0.083 minutes
(C) 12 minutes
(D) 165 minutes

30. At a steady percolation rate, a soil drains 360 mL in 9 minutes. How much water would drain in 15 minutes at the same rate?

(A) 216 mL
(B) 40 mL
(C) 366 mL
(D) 600 mL

31. Three columns of equal cross-section are tested. Sandy drains 400 mL in 5 min; loamy drains 400 mL in 10 min; clayey drains 400 mL in 20 min. Listing them from highest percolation rate to lowest gives:

(A) clayey, loamy, sandy
(B) loamy, sandy, clayey
(C) all equal, since each drained 400 mL
(D) sandy (80 mL/min), loamy (40 mL/min), clayey (20 mL/min)

32. A student measures: column 1 (deep) drains 500 mL in 10 min; column 2 (the same soil, half as deep) drains 500 mL in 5 min. The student concludes the second soil is 'more sandy'. The flaw is that:

(A) the deeper column must be sandier because it took longer
(B) the columns are the same soil, and the shorter time mainly reflects the smaller depth water must pass through
(C) depth has no effect on how long percolation takes
(D) a faster drain time always proves a coarser particle size

33. Equal 600 mL pours are tested on two soils. Soil A drains in 12 min; soil B drains in 8 min. Which statement is fully correct?

(A) soil A has the higher rate because 12 is greater than 8
(B) both have the same rate because the volume is equal

(C) soil A is more sandy because it took longer to drain

(D) soil B has the higher percolation rate (75 mL/min) and is the more sandy of the two

34. A percolation experiment must be a fair test. A student compares sandy and clayey soils but pours 300 mL on the sand and only 150 mL on the clay. The result is unreliable because:

(A) the soils should have been mixed together first

(B) the volume of water poured was not kept the same, so the comparison is unfair

(C) percolation rate cannot be measured for clay at all

(D) the timer should have been started after the water finished draining

35. Soil S drains 720 mL in 9 minutes; soil T drains 720 mL in 18 minutes. By what factor is soil S's percolation rate greater than soil T's?

(A) 2 times greater
(B) 9 times greater
(C) half as great
(D) the same

36. From their percolation data alone, you suspect soil X is loamy and soil Y is clayey. Soil X: 320 mL in 8 min; soil Y: 320 mL in 32 min. Which conclusion is sound?

(A) Y drains faster, consistent with Y being loam

(B) the rates are equal, so the soils must be identical

(C) X is clayey because it has the higher rate

(D) X's rate (40 mL/min) is four times Y's (10 mL/min), consistent with X being the more freely draining loam

37. The wearing away and removal of the fertile upper layer of soil by wind or running water is called soil erosion. Which scenario shows erosion (not merely percolation or weathering)?

(A) monsoon runoff carrying away the dark topsoil from a bare sloping field

(B) rainwater slowly seeping straight down into a level field

(C) freeze-thaw cracking a boulder into smaller fragments in place

(D) leaves rotting on the ground into dark humus

38. A bare ploughed hillside loses far more soil in the monsoon than a forested one. The clearest reason is that on the forested slope:

(A) plant roots bind the soil and the canopy and litter break the force of falling rain

(B) the forest soil is too heavy for any rain to move

(C) trees stop all rain from ever reaching the ground

(D) forests have no topsoil for the rain to remove

39. A hill farmer cuts level, step-like platforms across a slope and also ploughs his furrows along the contour lines instead of up and down. Together these two measures fight erosion mainly by:

(A) slowing the downhill rush of water so it soaks in instead of carrying soil away

(B) making the rainwater evaporate before it can move soil

(C) increasing the slope so water drains off faster

(D) removing all the soil deliberately before the rains

40. A dry, treeless plain suffers blowing dust and loses topsoil to the wind every summer. The most directly effective erosion-control measure here is to:

(A) dig the field deeper so the wind cannot reach the soil

(B) build terraces, which are meant for steep slopes

(C) plant rows of trees as shelter belts to slow the wind

(D) leave the field bare so the wind has nothing to catch

41. Why is the loss of topsoil to erosion treated as a far more serious problem than the loss of an equal depth of bedrock?

(A) topsoil is the humus-rich, nutrient-bearing layer crops feed from, and it takes a very long time to rebuild

(B) bedrock is the only layer that holds nutrients for crops

(C) topsoil grows back fully within a single season

(D) topsoil has no role in farming, so its loss is harmless

42. A factory dumps plastic waste and toxic chemicals onto nearby fields, and over years crops there fail. This is best described as:

(A) soil erosion, because the chemicals wash the topsoil away

(B) weathering, because chemicals break rock into soil

(C) healthy humus formation that improves fertility

(D) soil pollution, which can poison soil organisms and make the soil unfit for crops

43. A student heats a known mass of garden soil strongly in an open dish until no more change is seen, then re-weighs it; the mass falls. After cooling, the soil also looks paler. The mass loss is mainly because heating:

(A) destroyed the mineral sand and silt particles

(B) added air to the soil, which weighs nothing(C) turned the soil into bedrock

(D) drove off the soil's moisture and burnt away its humus

44. Two soils are heated to drive off and weigh their water content; the clayey soil loses far more mass as water than the sandy soil. This is consistent with the fact that clayey soil:

(A) has the higher percolation rate, so it holds more water

(B) is made of larger particles that trap more water

(C) has a higher water-holding capacity, so it retains more moisture

(D) contains no air spaces, so all of it must be water

45. Earthworms are often called 'farmers' friends'. Considering their effect on soil structure, they help mainly by:

(A) burrowing and mixing the soil, which improves aeration and lets water percolate better

(B) compacting the soil so its pores close up

(C) removing all the humus so the soil stays clean

(D) turning sandy soil into solid clay

46. A profile shows a very thin A horizon over a thick, pale, fast-draining sandy C-like layer, and crops there grow poorly and dry out. The most likely combined cause is:

(A) too much humus making the soil water-logged

(B) a thick clay subsoil holding too much water around the roots

(C) high water-holding capacity drying the crops out

(D) little humus in a thin topsoil plus fast percolation that drains water beyond the roots

47. A gardener over-waters a pot of clayey soil every day and the plant wilts and rots. Explaining this to a friend, the correct chain of reasoning is:

(A) clay drains the water away so fast that the roots dry out and wilt

(B) clay holds the extra water in its tiny pores, the pores stay full, air is shut out, and the roots suffocate

(C) the extra water turns the clay into sand and the roots fall out

(D) clay has so much air already that adding water adds even more air

48. On a steep hillside, a farmer ploughs straight up and down the slope and clears the remaining trees, then the monsoon cuts deep gullies and removes much soil. The single best corrective set of actions is to:

(A) plough even more steeply up and down to speed drainage

(B) terrace the slope, plough along the contour, and replant vegetation to bind the soil

(C) clear any vegetation that is left so water runs off faster

(D) pour extra water down the slope to wash the gullies smooth

49. Two adjacent fields drain the same 400 mL of water: the well-managed field with added humus takes 10 min, while the over-cultivated, compacted field takes 25 min. The added humus and looser structure most directly explain why the managed field has the:

(A) lower percolation rate, because humus blocks the pores

(B) same rate, because both drained 400 mL

(C) higher percolation rate (40 mL/min versus 16 mL/min) because of more open pore spaces

(D) higher rate only because its particles are larger grains of sand

50. A profile dug under an old grassland shows a deep, dark A horizon; a profile under a long-cultivated nearby field shows a thin, pale A horizon. The single best explanation tying horizon and management together is that:

(A) continuous cropping and exposure removed organic matter and let topsoil erode, thinning the A horizon
(B) cultivation made the C horizon rise up and replace the A horizon
(C) grassland soils have no humus, which is why they look dark
(D) the cultivated field simply has a thicker bedrock

51. A student ranks three soils by percolation rate using equal volumes and finds clayey slowest and sandy fastest, then claims 'therefore clayey soil is useless for farming.' The best correction is that:
(A) a slow percolation rate is exactly what suits water-loving crops like paddy, so clay is valuable for the right crop
(B) clay's slow rate means it can grow no crop at all
(C) the slowest soil always grows the most crops of every kind
(D) percolation rate has nothing to do with which crops suit a soil

52. Soil 1 drains 540 mL in 9 min; soil 2 drains 540 mL in 27 min; soil 3 drains 540 mL in 6 min. Ordering them as sandy, loamy, clayey by their likely texture gives:
(A) soil 3 sandy, soil 1 loamy, soil 2 clayey(B) soil 2 sandy, soil 1 loamy, soil 3 clayey
(C) soil 1 sandy, soil 3 loamy, soil 2 clayey
(D) all three the same texture, since each drained 540 mL

53. A farmer measures percolation, then plans crops: field A = 5 mL/min, field B = 70 mL/min. The most sensible matching of crops is:
(A) paddy in field B because fast drainage suits rice
(B) paddy in field A (slow drainage holds water) and groundnut in field B (fast drainage, loose soil)
(C) groundnut in field A because rice-like wetness suits groundnut
(D) the same crop in both, since percolation does not affect crop choice

54. A claim states: 'Because sandy soil has the highest percolation rate, it must also have the highest water-holding capacity.' This is a misconception because percolation rate and water-holding capacity are:
(A) inversely related — sandy drains fast but holds little, clay drains slow but holds much
(B) directly related, so the fastest-draining soil also holds the most water
(C) identical properties, so they are always equal
(D) both highest in sandy soil at the same time

55. A soil-conservation officer recommends covering bare fields between crops with a layer of straw mulch. The chief erosion-fighting benefit of the mulch is that it:
(A) makes the soil so heavy that wind and water cannot move it
(B) seals the soil so no rain can ever percolate in

(C) shields the soil surface from rain impact and wind, and holds moisture, so topsoil stays in place

(D) turns the topsoil into bedrock that cannot erode

56. During a percolation experiment, water poured onto a clayey sample sits on top for a long time before slowly seeping down, while on sandy soil it vanishes almost at once. The clay's behaviour is best explained by:

(A) its large particles leaving wide channels for slow flow

(B) its tiny, tightly packed pores resisting the downward flow of water

(C) the water being absorbed and destroyed by the clay

(D) clay having a higher percolation rate than sand

57. A soil sample drains 240 mL in the first 6 minutes at a steady rate. Assuming the same rate continues, the total volume drained after 15 minutes from the start is:

(A) 360 mL (B) 600 mL

(C) 240 mL (D) 150 mL

58. Reasoning about a soil profile, which statement is correct?

(A) the bedrock is the layer richest in humus and roots

(B) the B horizon is the surface layer where leaves first fall

(C) the A horizon usually has the most humus and supports most plant roots, while bedrock has none

(D) the C horizon contains more humus than the A horizon

59. If you wanted to take a soil sample richest in humus and active plant roots, and a second sample of barely-weathered parent rock fragments, you should dig in the:

(A) C horizon for the humus-rich sample and the A horizon for the parent-rock fragments

(B) bedrock for both samples, since it holds everything

(C) B horizon for both, because the subsoil contains the most humus

(D) A horizon for the humus-rich sample and the C horizon for the parent-rock fragments

60. Two columns of the same depth are compared. Column 1 drains 450 mL in 9 minutes; column 2 drains 450 mL in 15 minutes. By how many mL/min does column 1's percolation rate exceed column 2's?

(A) 6 mL/min (B) 70 mL/min

(C) 0 mL/min (D) 20 mL/min

Solutions — Science (Class 7), Chapter 9: Soil — Paper 3

Paper 3 — (progressively harder)

Marking: **+4** correct, **−1** wrong, **0** unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	11	A	21	C	31	D	41	A	51	A
2	C	12	B	22	A	32	B	42	D	52	A
3	A	13	C	23	C	33	D	43	D	53	B
4	B	14	D	24	A	34	B	44	C	54	A
5	D	15	D	25	B	35	A	45	A	55	C
6	A	16	C	26	D	36	D	46	D	56	B
7	D	17	B	27	D	37	A	47	B	57	B
8	B	18	B	28	B	38	A	48	B	58	C
9	C	19	C	29	C	39	A	49	C	59	D
10	B	20	C	30	D	40	C	50	A	60	D

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (C) The spongy leaf layer is the O (organic) horizon, the dark crumbly band is the humus-rich A (topsoil), the paler band where clay/iron accumulate is the B (subsoil), loose rock fragments are the C horizon, and the solid layer is bedrock — so O, A, B, C, bedrock. Swapping A and B (O, B, A, C) is wrong because the dark humus band sits directly under the litter, above the wash-down layer. Putting C above B (A, O, C, B) both reorders the surface layers and inverts subsoil/weathered-rock. Listing bedrock first describes bottom-to-top, the reverse of what was asked.

2. (C) Water percolating down through the topsoil (A) dissolves and washes fine clay and minerals out of A and deposits them lower down in B — that is why B is the zone of accumulation and A is sometimes called the zone of leaching. Solid bedrock does not push minerals upward into B; weathering of bedrock feeds the C horizon, not B by capillary rise of finished minerals. Wind moves loose surface dust, not material into a buried subsoil layer. Earthworms mainly mix the upper layers; they do not transport the C horizon's material up into B as the chief source of accumulation.

3. (A) A thick dark A horizon means lots of humus, which builds up where plant litter keeps falling and decaying and where erosion is not carrying topsoil away — exactly the forest case. Root action does crack rock, but that feeds the deep C horizon over very long times and does not explain a darker, humus-rich topsoil. A thick B horizon does not 'push up' the A horizon; the layers form in place from the surface down. Faster drainage actually washes minerals downward out of the topsoil, which would not enrich the A horizon.

4. (B) The C horizon is the partly broken-down parent rock just above the solid bedrock, with little organic matter — so describing it as looser, weathered pieces of the rock below is broadly right. It is not the humus-rich layer; that is the A horizon near the surface. C lies near the bottom of the profile, well below the topsoil, not above it. And it is rock material, not a layer made only of living organisms.

5. (D) Topsoil increasing only over centuries and millennia shows that soil formation by weathering plus humus build-up is a very slow process. It is not produced within a few rainstorms — that contradicts the near-zero reading at 50 years. A single year's rainfall does not set the total depth; the trend tracks elapsed time, not one season. And soil clearly gets deeper, not thinner, with age, so 'deepest first' is backwards.

6. (A) Weathering is the combined slow work of water that freezes and expands in cracks, wind that scours surfaces, and roots that wedge into and split rock — all over long times. A single earthquake shifts and fractures rock but does not by itself convert it to soil; soil still needs prolonged weathering and humus. Bedrock's own weight does not powder itself. Humus forms from decay and enriches soil; it does not assemble rock, so it cannot be the agent that breaks rock down.

7. (D) Weathering generally goes faster where it is both warm and wet, because water drives chemical breakdown and freeze-thaw/physical action, and warmth accelerates reactions — hence the deep tropical soil. Cold dry conditions slow weathering, so they cannot be the faster route. The contrast is climate-driven, not because tropical rock is inherently softer. And high altitude does not by itself make deeper soil — the bare mountaintop is the counterexample.

8. (B) Roots need oxygen from the air spaces in soil to respire, so a soil with no air directly suffocates them — this is why water-logged soil harms plants. Minerals would still be present in the particles, so nutrient access is not the most direct loss. Water would still be present (in fact a no-air soil is usually water-logged), so it is not that roots cannot take up any water. Roots do not turn soil into humus; humus comes from decaying organic matter over time.

9. (C) Tiny tightly-packed particles are characteristic of clayey soil (P), large loose grains are sandy soil (Q), and a balanced mix of sizes plus humus is loamy soil (R) — so clayey, sandy, loamy. Calling P sandy reverses the small/large particle sizes. 'Loamy, clayey, sandy' mislabels the tiny-particle sample as loamy. 'Clayey, loamy, sandy' wrongly tags the large-grain sample as loamy and the mixed sample as sandy.

10. (B) Clay percolates slowly precisely because its particles are tiny and tightly packed, so the pore spaces are minute and water seeps through slowly — the opposite of the student's 'large particles, big gaps' claim. Large gaps and fast drainage describe sandy soil, not clay. Clay particles are far smaller than sand grains, not the same size. And clay actually holds water very well rather than being unable to retain it.

11. (A) Sandy soil has large pores, so it lets water drain quickly and retains little — pot A drains more and holds less than the clayey pot B. The reverse (more retained, less drained) describes clayey soil B, not sandy A. The two soils differ in pore size, so their drainage cannot be identical. And sand does not block water; its big gaps let water through fastest, the very opposite of 'no drainage'.

12. (B) The soil that holds the most water (clayey) has tiny, tightly packed pores, and those same small pores make water seep through slowly — so highest retention pairs with the slowest, not fastest, percolation. They do not rise together, so the first option is false. They are different, inversely related properties, not one property measured twice. And no single soil maximises both at once; loamy is a moderate compromise, not a soil with both the very highest retention and the very fastest drainage.

13. (C) When clay's tiny pores fill with water and stay full (water-logged), almost no air is left for roots, so aeration is poorest there. Loose sandy soil keeps large air gaps and drains fast, so it is well aerated. Well-drained loam holds some water but its extra drains away, leaving air, so it is reasonably aerated. Dry sandy soil is full of air, the opposite of poorly aerated.

14. (D) Loam's mix of particle sizes plus humus lets it retain enough moisture for roots while draining the excess and keeping air spaces — a balance that suits most crops. It does not drain faster than every soil; sandy drains fastest. It does not hold more water than every soil; clay holds the most. And loam is not made only of large particles — it is a mixture, so 'impossible to compact' is false.

15. (D) Sandy soil drains and dries fastest, clay holds water and stays soggy longest, and loam falls in between — matching the first option. Clay does not drain fastest; its tiny pores drain slowest. Sand does not hold the most water; it holds the least, so it cannot stay soggy longest. And because the soils differ in pore size, they cannot dry at exactly the same time.

16. (C) Humus holds moisture and supplies nutrients, so adding it to fast-draining sand boosts water retention and fertility while drainage stays workable — the practical fix gardeners use. It does not transmute sand into airless clay; it is organic matter, not clay particles. It does not seal the soil against all percolation. And humus actually supports more soil organisms, not fewer, so 'removes all organisms' is wrong.

17. (B) Loosening the soil opens up the gaps (pores) between particles, so water percolates in more easily; compaction squeezes those pores shut and slows it. Ploughing does not change the particles into all-large grains — it only rearranges them. Packing does not add humus, and compaction slows rather than speeds drainage. Compacted soil has smaller, not bigger, pores than loose soil, which is exactly why it soaks in less.

18. (B) Soil organisms thrive where there is moisture, organic food (humus) and air — all abundant in the loamy patch and scarce in the salty, dry sand. Temperature is not the deciding factor here, and colder is not automatically more living. Large particles (sand) actually offer less retained moisture and food, so they are not preferred. Very fast drainage dries the soil and removes moisture, which would reduce, not boost, the organism count.

19. (C) Paddy needs water held around the roots, which water-retentive clay provides, while groundnut needs the easy drainage and looseness of sandy or sandy-loam soil — so clay for paddy, sandy-loam for groundnut. Reversing them (sand for paddy, clay for groundnut) drains the rice field and water-logs the groundnut. Clay for both would water-log the groundnut. Gravel cannot hold flood water for paddy, and water-logged clay suffocates groundnut, so the last pairing fails on both counts.

20. (C) Paddy thrives in moist, water-holding soil and is the classic match for heavy clay. A cactus needs fast-draining, dry soil and would rot in moist clay. Groundnut needs loose, well-drained ground, not water-retentive clay. Watermelon on a sandy ridge describes a sandy, well-drained setting — the opposite of moist clay.

21. (C) Clayey-to-loamy, moisture-holding soil suits wheat and gram, so those are the right picks. Cotton actually prefers lighter, well-drained sandy-loam, not water-logged clay, so that reasoning is wrong. Crops needing pure sand would suffer in a moisture-retentive clay-loam. And clayey-loam is in fact fertile and supports many crops, so 'no crop' is false.

22. (A) Sandy-loam is light and well-drained, which suits lentils (and cotton). Paddy needs standing water, so fast drainage is exactly wrong for rice. Soil type strongly affects which crop will thrive, so 'makes no difference' is false. Crops needing flooded fields would fail on free-draining sandy-loam.

23. (C) Keeping clay flooded fills its pores with water and leaves no air for the roots to breathe, harming a dry-land crop like gram — so 'more water always helps' is a misconception. Flooding clay does not make it drain fast or dry out; clay drains slowly and stays wet. Adding water does not convert clay into sand. And gram is not a water plant; it needs well-aerated, moist-not-flooded soil.

24. (A) Vegetables wanting steady moisture but no water-logging are best served by soil M's balance of moderate holding plus drainage of the excess (a loamy character). Soil N's strong retention and very slow drainage risks water-logging, and 'more water is always better' is the misconception being tested. Slow drainage actually reduces air to roots, not increases it. And soil M does hold some water — it is not a zero-retention soil — so the last option misdescribes it.

25. (B) Rate = volume ÷ time = 250 mL ÷ 5 min = 50 mL/min. Multiplying 250 × 5 gives 1250, the wrong operation. Dividing time by volume (5 ÷ 250) gives 0.02, the inverted ratio. Subtracting (250 − 5 = 245) is not a rate at all; the definition is a quotient, not a difference.

26. (D) Rate = 90 mL ÷ 30 min = 3 mL/min, a suitably low value for slow-draining clay. Multiplying 90 × 30 gives 2700, the wrong operation. Inverting to 30 ÷ 90 gives about 0.33, the time-over-volume mistake. Subtracting 90 − 30 = 60 is a difference, not a rate.

27. (D) Rate of X = 200 ÷ 4 = 50 mL/min; rate of Y = 200 ÷ 25 = 8 mL/min; ratio 50 : 8 = 25 : 4. The reverse 4 : 25 mistakenly uses the times in the same order as the rates, but the faster soil (smaller time) has the bigger rate. They are not equal (1 : 1), since X drains much faster. And 200 : 200 just compares the equal volumes, ignoring time, which is irrelevant to the rate ratio.

28. (B) Rate P = 300 ÷ 6 = 50 mL/min; rate Q = 480 ÷ 12 = 40 mL/min, so P is faster. Comparing only the larger volume (Q's 480 mL) ignores that Q also took twice as long, so volume alone is misleading. The two rates are 50 vs 40, not equal. Taking the longer time makes a soil slower, not faster, so Q's longer time does not make it higher-rate.

29. (C) Since rate = volume ÷ time, time = volume ÷ rate = 180 ÷ 15 = 12 minutes. Multiplying 180 × 15 gives 2700, the wrong operation. Inverting to 15 ÷ 180 gives about 0.083, dividing the wrong way. Subtracting 180 − 15 = 165 treats a rate problem as a difference, which it is not.

30. (D) Rate = 360 ÷ 9 = 40 mL/min; in 15 minutes that is 40 × 15 = 600 mL. The value 216 comes from 360 ÷ 15 × 9 done in the wrong order, undercounting. 40 mL is just the per-minute rate, not the 15-minute total. 366 (= 360 + 6) treats the change as an addition rather than scaling with time.

31. (D) Equal volumes mean the shortest time has the highest rate: sandy 400÷5 = 80, loamy 400÷10 = 40, clayey 400÷20 = 20 mL/min, so sandy > loamy > clayey. The reverse 'clayey, loamy, sandy' wrongly ranks the slowest soil highest. 'Loamy, sandy, clayey' misorders the top two. They are not all equal — the equal volume only highlights that the differing times set the rates.

32. (B) Because both columns are the same soil, the difference in time comes from depth — water passes a shorter distance in the half-depth column — not from a difference in particle size, so calling the second 'more sandy' is the error. Longer time in the deeper column reflects depth, not greater sandiness, so the deep one is not sandier. Depth clearly does affect the time, so 'no effect' is false. And a faster drain time does not by itself prove coarser particles when the variable changed is depth, not soil type.

33. (D) Rate A = 600 ÷ 12 = 50 mL/min; rate B = 600 ÷ 8 = 75 mL/min, so B is faster and the more sandy. A larger time means a lower, not higher, rate, so A is not higher just because 12 > 8. Equal volume does not make the rates equal when the times differ. And the soil that took longer (A) is the slower, more clayey one, not the more sandy one.

34. (B) A fair test keeps everything the same except the soil; pouring different volumes (300 vs 150 mL) confounds the comparison. Mixing the soils together would destroy the very comparison being made, so that is not the fix. Clay's percolation rate can be measured — it is simply low — so that claim is false. The timer must run while the water drains, not after it finishes, so that procedure is also wrong.

35. (A) Rate S = 720 ÷ 9 = 80 mL/min; rate T = 720 ÷ 18 = 40 mL/min; 80 ÷ 40 = 2, so S is twice as fast. It is not 9 times — 9 is just one of the times, not the ratio. S is faster, not half as great, since it drains the same volume in less time. And the rates are not the same because the times differ.

36. (D) Rate X = 320 ÷ 8 = 40 mL/min and rate Y = 320 ÷ 32 = 10 mL/min, so X drains four times faster, matching the freer-draining loam. Y has the lower rate, so it does not drain faster, and clay-like Y is not the loam. The rates (40 vs 10) are not equal, so the soils are not identical. The higher-rate soil is the more freely draining one (loam), not clayey, so calling X clayey is backwards.

37. (A) Erosion is the removal and transport of topsoil, exactly what runoff carrying soil off a bare slope shows. Rainwater seeping straight down is percolation, which moves water, not topsoil, away. Freeze-thaw cracking a boulder in place is weathering — it breaks rock but does not carry topsoil off. Leaves rotting into humus is decomposition that builds soil, the opposite of removing it.

38. (A) Roots hold soil particles together and the leaves, branches and litter cushion the impact of rain, so far less topsoil is washed off a forested slope. The protection is not that forest soil is too heavy — soil weight is similar. Trees intercept some rain but do not stop all of it, so 'no rain reaches the ground' is false. Forests in fact build deep, humus-rich topsoil, so 'no topsoil to remove' is wrong.

39. (A) Terracing and contour ploughing both create level catchments that slow runoff, giving water time to soak in rather than sheet downhill and strip topsoil. They do not work by evaporating the rain. They reduce the effective slope and water speed, not increase the slope or drainage speed. And they certainly do not remove the soil on purpose — their whole aim is to keep it.

40. (C) Shelter belts of trees slow the wind near the ground and cut wind erosion on flat, exposed plains. Digging deeper does not stop wind from lifting surface soil. Terraces are designed for water erosion on slopes, not wind erosion on a flat plain. Leaving the field bare removes the protective cover and makes wind erosion worse, not better.

41. (A) Topsoil is the fertile, humus-rich layer where crops get their nutrients, and because it forms only over very long times, its loss is hard to reverse and seriously harms farming. Bedrock is not the nutrient-holding layer — the topsoil is. Topsoil does not regrow in a single season; it builds over centuries. And topsoil is central, not irrelevant, to farming, so its loss is far from harmless.

42. (D) Adding plastic and toxic chemicals contaminates the soil — soil pollution — harming the organisms and making it unfit for crops. It is not erosion, which is the physical removal of topsoil, not contamination. It is not weathering; toxic dumping does not constitute the helpful breakdown of rock into soil. And it is the opposite of humus formation, which enriches soil rather than poisoning it.

43. (D) Strong heating evaporates the soil's water and burns off the dark organic humus, which both reduces the mass and lightens the colour. Mineral sand and silt are not destroyed by such heating, so they are not the source of the loss. Adding air would not reduce mass, and the soil is losing matter, not gaining it. Heating loose soil does not fuse it into bedrock.

44. (C) Clay's tiny pores give it a high water-holding capacity, so a sample retains more moisture and loses more mass as water on heating. A high percolation rate means water drains away fast — that would leave less water held, so it cannot explain more retention. Clay has smaller, not larger, particles; small particles and tight packing are what hold the water. And clay does contain some air; the loss is water, not the absence of all air spaces.

45. (A) Earthworm burrows loosen and mix the soil, opening channels that let in air and let water percolate — improving both aeration and drainage. They loosen rather than compact, so they do not close up the pores. They feed on and help recycle organic matter into humus, enriching rather than removing it. And they do not transform sandy soil into clay; they do not change particle type.

46. (D) A thin, humus-poor A horizon offers few nutrients, and the underlying coarse, fast-draining material lets water rush past the root zone, so crops starve and dry out — both factors point the same way. There is little humus, so 'too much humus and water-logging' contradicts the description. The layer is sandy, not a water-holding clay subsoil. And high water-holding capacity would keep soil wet, which cannot be the cause of drying out.

47. (B) Clay's tiny pores retain the surplus water, so they remain full, air is excluded, and the roots cannot breathe — leading to rot and wilting. Clay does not drain fast; the problem is too little drainage, not too much. Water does not convert clay to sand. And water-logged clay loses air rather than gaining it, so 'adds even more air' is exactly backwards.

48. (B) Terracing plus contour ploughing slows runoff and replanted roots bind the soil, directly countering the gullying. Ploughing up and down more steeply channels water faster and worsens erosion. Clearing the remaining vegetation removes the very cover that protects the soil. Pouring extra water down the slope adds to the runoff that is already cutting gullies, making the damage worse.

49. (C) Managed rate = 400 ÷ 10 = 40 mL/min; compacted rate = 400 ÷ 25 = 16 mL/min, so the managed field percolates faster, thanks to humus and looser structure keeping pores open. Humus and loosening open pores rather than block them, so the rate is higher, not lower. Equal volume does not make the rates equal when the times differ. And the improvement comes from structure and pore space, not from converting the soil to coarse sand.

50. (A) Years of cropping with little organic return, plus exposure to erosion, strip humus and topsoil, so the cultivated field's A horizon is thin and pale, while undisturbed grassland keeps a deep dark humus layer. The C horizon does not rise to replace the A; horizons form from the surface down. Grassland soils are dark precisely because they are rich in humus, so 'no humus' is false. Bedrock thickness is unrelated to how thick the surface A horizon is.

51. (A) Clay's slow percolation and high water retention make it ideal for water-loving crops such as paddy, so it is valuable for the right crop, not useless. It is not true that clay grows no crop — it grows paddy well. Nor is the slowest soil best for every crop; dry-land crops prefer free-draining soils. And percolation rate is closely tied to crop suitability — it helps decide which crop a soil fits, so it is far from irrelevant.

52. (A) Rates: soil 3 = 540÷6 = 90, soil 1 = 540÷9 = 60, soil 2 = 540÷27 = 20 mL/min; fastest is sandiest, slowest is clayiest, so soil 3 sandy, soil 1 loamy, soil 2 clayey. Calling soil 2 (slowest) sandy is backwards. Calling soil 1 sandy puts the second-fastest, not the fastest, as sand. And equal volume does not make the textures the same — the differing times reveal different textures.

53. (B) Field A's very slow percolation (5 mL/min) means it holds water — ideal for paddy — while field B's fast rate (70 mL/min) signals loose, free-draining soil that suits groundnut. Putting paddy in the fast-draining field B would leave rice short of standing water. Groundnut needs dry, well-drained soil, not the wet, slow-draining field A. And percolation strongly affects crop choice, so planting the same crop in both ignores the data.

54. (A) High percolation goes with low retention and vice versa, so sandy (fast-draining) holds the least water while clay (slow-draining) holds the most — they move oppositely. They are not directly related; the fastest-draining soil holds the least, not the most, water. They are different properties, not identical, so they are not always equal. And sandy soil is not highest in both at once — it is highest in percolation but lowest in retention.

55. (C) Mulch cushions the soil against the beating of rain and the lifting force of wind and conserves moisture, so the topsoil is not dislodged. It does not work by making soil too heavy to move — its protection is a cover, not added weight. It does not seal the soil; water still percolates through a mulched surface. And it does not turn topsoil into bedrock; it simply protects the loose topsoil that is there.

56. (B) Clay's minute, closely packed pores offer high resistance to flow, so water pools and seeps down slowly. Clay has small, not large, particles, so 'wide channels' is wrong. The water is not destroyed; it is held in the pores and eventually drains. And clay's percolation rate is lower than sand's, not higher, which is exactly why water lingers on clay but vanishes into sand.

57. (B) Rate = 240 ÷ 6 = 40 mL/min; over 15 minutes total = 40 × 15 = 600 mL. The figure 360 mL is the amount for just the extra 9 minutes (40 × 9), not the total from the start. 240 mL is only the first 6 minutes' drainage. 150 mL ignores the rate and seems drawn from the 15-minute number alone, which is a time, not a volume.

58. (C) Humus and the bulk of plant roots are concentrated in the dark A horizon (topsoil), while solid bedrock at the base has essentially no humus or roots. Bedrock is the least, not the most, humus-rich layer. The surface layer where leaves fall is the O horizon, not the buried B horizon. And the C horizon (weathered parent rock) has far less humus than the A horizon, not more.

59. (D) The humus-rich, root-filled layer is the A horizon (topsoil), while loose, barely-weathered parent-rock fragments are found in the C horizon just above bedrock — so A for humus, C for rock fragments. Swapping them is backwards: C is rock-rich and humus-poor, A is humus-rich. Solid bedrock is unweathered rock with no humus, so it cannot supply the humus sample. And the B horizon is the wash-down subsoil, not the layer with the most humus, so it fits neither sample.

60. (D) Rate 1 = 450 ÷ 9 = 50 mL/min; rate 2 = 450 ÷ 15 = 30 mL/min; the difference is 50 − 30 = 20 mL/min. The value 6 is just the difference of the two times (15 − 9), not of the rates. 70 wrongly adds the two times (9 + 15) into mL/min. And the rates are not equal, so the difference is not 0 — column 1 is the faster of the two.